

1. "What happens when you type www.google.com and press Enter?"

When we type www.google.com into our browser's address bar and press **Enter**, it seems like the Google homepage appears instantly. But in the background, our computer, browser, network, and Google's servers perform a long chain of operations in just milliseconds.

DNS (Domain Name System) is a system that translates human-friendly website names like www.google.com into computer-friendly IP addresses like **142.250.190.14**.

Similarly:

- we type www.google.com (the name).
- The internet needs an IP address (the number) to reach Google's server.
- DNS acts like a phonebook that finds the number for the name.

2. How does your computer find Google's IP address?

When we type www.google.com into our browser, our computer doesn't immediately know where Google is located. It must find the IP address (example: 142.250.190.14) so it can contact the correct server.

3. What Is an IP Address?

An IP address (Internet Protocol address) is a unique number assigned to every device on a network. It works like a home address for devices on the internet so they can find, identify, and communicate with each other.

Why Do We Need IP Addresses?

Because without IP addresses:

- Devices wouldn't know where to send data
- Websites couldn't be found
- The internet simply wouldn't work

4. **What is TCP?**

TCP (Transmission Control Protocol) is one of the main communication protocols that the internet uses to send data reliably between two devices.

We need TCP for the following reasons

1. Reliability

2. Ordered Delivery
3. Error Checking
4. Flow Control

5. What Is the TCP Handshake? (Simple Explanation)

TCP handshake like two people introducing themselves before they start a conversation. Before sending data, your computer and the server must make sure both sides are ready. So they do a **3-step greeting**, just like polite communication.

The TCP handshake is a polite “hello” exchange between your computer and a server to make sure both are ready before talking

6. What Is a Port Number?

A **port number** is like a **door** on a computer that allows specific types of internet traffic to enter or leave.

Your device has one IP address, but it uses **different port numbers** to handle different services—just like a house has one address but many doors for different rooms.

Service	Port Number	Use
HTTP	80	Unsecured website traffic
HTTPS	443	Secure website traffic (like Google)
FTP	21	File transfer
SSH	22	Secure remote login
SMTP	25	Email sending

7. What Is HTTP?

HTTP (HyperText Transfer Protocol) is the rules or instructions that computers use to communicate on the web.

It decides:

- How your browser asks for a webpage
- How the server sends the webpage back

8. What Does an HTTP Request Look Like? (Simple Example)

GET / HTTP/1.1

Host: www.google.com

GET → The method (browser is asking for a page).

/ → The homepage of the website.

HTTP/1.1 → The version of HTTP being used.

Host: www.google.com → Tells which website you want (useful when many sites share one server).

9. What information does your browser send?

When your browser sends a request to a website, it includes several details to help the server understand what you want.

Your browser sends:

- **The method** (GET/POST)
- **The path (URL)**
- **Website name (Host header)**
- **Your browser type** (User-Agent: Chrome, Firefox, Safari...)
- **What formats you accept** (HTML, images, compressed data)
- **Language preference**
- **Cookies** (login info, settings)
- **Referrer** (page you came from, sometimes)

10. What does the server send back?

The server replies with an **HTTP response** containing:

- **Status code** (200 = OK, 404 = Not Found)
- **Headers** (content type, length, caching instructions)
- **HTML page**
- **CSS files**
- **JavaScript files**
- **Images, fonts, videos** (if required)

11. Explain DNS as the , where computers look up where to deliver email.

DNS (Domain Name System) works like the **internet's phonebook**.

Just like you look up a person's name to find their phone number, **computers look up a domain name to find out where to deliver email**.

When you send an email to someone like:

john@example.com

What mail server handles email for example.com?"

DNS then answers with the mail server's address — usually called **MX records** (Mail Exchange records).

So in simple terms:

DNS = the phonebook that tells computers which mail server to send an email to.

12. Describe and how DNS tells your computer which server handles email for a domain.

DNS uses special records called MX (Mail Exchange) records to tell your computer which mail server is responsible for receiving email for a specific domain.

When you send an email, your computer asks DNS for the domain's MX record, and DNS responds with the name of the mail server that should receive the message.

DNS Replies with the Mail Server's Address

DNS will respond with:

example.com → MX → mail.example.com

mail.example.com → IP → 142.250.27.45

13. Introduce SMTP as the protocol responsible for , like the postal service of the internet.

SMTP (Simple Mail Transfer Protocol) is the protocol responsible for sending emails from one server to another, just like the postal service of the internet.

It collects your email, carries it across the network, and delivers it to the correct mail server—just like a post office sends letters to the right destination.

TCP (Transmission Control Protocol) makes sure your email is delivered safely, correctly, and in the right order.

While SMTP is the mailman, **TCP is the road system and safety rules** that guarantee:

- No parts of the email get lost
- The message arrives exactly as it was sent
- Any missing data is retransmitted
- Both computers stay connected during the transfer

14. Use an analogy such as “introducing yourself before starting a conversation.”

Explain ports as “doors” that servers listen on.

Mention common SMTP ports: .

Before two computers start sending an email, they perform a **TCP handshake**. This is like two people politely introducing themselves before they talk.

Example:

- **Computer A:** “Hello, can we talk?”
- **Computer B:** “Yes, I can hear you. Can you hear me?”
- **Computer A:** “Yes. Let’s begin.”

A **port** is like a **door** on a computer that lets different services communicate.

- A server has many “doors.”
- Each door is assigned to a specific type of traffic.
- Email uses one door, web browsing uses another, etc.

So when SMTP wants to send an email, it goes to the server’s **email door**.

Common SMTP Ports

SMTP uses specific ports (doors) for sending mail:

- **Port 25** → Server-to-server email delivery
- **Port 465** → SMTP over SSL (secure)
- **Port 587** → Modern, secure mail submission (used by email apps)

15. Describe that the email includes:

-Sender’s address

from: alice@example.com

-Recipient’s address

to: bob@example.com

-Subject

Subject: Meeting Tomorrow

-Message body

Hi Bob,

Just reminding you about tomorrow’s meeting at 10 AM.

Attachments (via MIME)

they are added using a system called **MIME (Multipurpose Internet Mail Extensions)**.

MIME tells the email server:

- What type of file is attached
- How to encode it
- How to deliver it safely

Example:

- image.jpg
- resume.pdf
- report.docx

In short:

An email contains:

- Who it's from
- Who it's going to
- What it's about
- What it says
- Any files attached (handled by MIME)

16 . Provide a simple example of commands like

HELLO

Introduces the sending server.

“Hello, this is sender-mail-server.com.”

MAIL FROM:

Tells the receiving server who the email is from.

“The sender is alice@example.com

RCPT TO:

Specifies who the email should be delivered to.

“The recipient is bob@example.com

DATA

Starts the actual message content (subject + body).

The message ends with a single period (.) on a line by itself.

17.Explain metadata such as time sent, mail headers, and message content.

1. Time Sent

This tells:

- The exact date and time the email was sent
- The time zone
- Sometimes even the server that handled it

Example:

Date: Tue, 19 Nov 2025 10:45:23 +0530

2. Mail Headers

Headers contain important technical details about the email.

Some headers you might see:

- From: who sent the email
- To: who receives it
- Subject: what the email is about
- Message-ID: a unique ID for the email
- Received: the path the email took through different servers
- Content-Type: tells if the message is plain text or HTML
- MIME-Version: for attachments

Example:

From: alice@example.com

To: bob@example.com

Subject: Meeting Reminder

3. Message Content

This is the actual **body of the email** — the text you wrote.

It may include:

- Plain text
- HTML formatting
- Signatures
- Embedded images

Example:

Hi Bob,

Don't forget our meeting today at 4 PM.

18. Explain how the email moves between servers across the internet.

1. Your Email App Sends the Message to Your Outgoing Mail Server

Your app (Gmail, Outlook, Yahoo, etc.) sends the email to your **SMTP server**.

Example:

alice@example.com sends through smtp.example.com

This server is responsible for starting the delivery process.

2. Your SMTP Server Looks Up the Receiver's Mail Server Using DNS

It asks DNS:

“Where should I deliver mail for **bob@anotherdomain.com**?”

DNS replies with the destination **MX record** →
something like mail.anotherdomain.com

3. Your Server Connects to the Recipient's Server Using SMTP

Your server opens a connection (like calling another post office):

- It introduces itself
- It says who the email is from
- It says who it's for
- It transfers the message

This is the **SMTP conversation**.

4. If the Recipient's Server Is Far Away, the Email May Travel Through Multiple Servers

Sometimes your server cannot reach the receiver directly.

In that case, it hands off the message to **intermediate mail servers** (relay servers).

These servers act like transfer hubs — similar to postal sorting centers.

The email hops across the internet until it reaches the final server.

19. Describe how the server: Accepts the message Checks for spam Stores it in the user's mailbox

The Server Accepts the Message

When the sending server connects, the receiving server:

- Confirms the recipient's email address exists
- Listens for the SMTP commands (HELO, MAIL FROM, RCPT TO, DATA)
- Receives the full message content (headers, body, attachments)

If everything is valid, the mail server replies with:

250 OK: Message accepted

This means the server has officially taken responsibility for delivering the email.

The Server Checks for Spam

Before storing the message, the server runs **spam and security checks**, such as:

SPF

Confirms whether the sender's mail server is allowed to send mail for that domain.

DKIM

Verifies that the message hasn't been changed during transit.

DMARC

Checks the sender's identity and applies the domain's security policies.

Spam Filters

Looks for:

- Suspicious keywords
- Blacklisted IP addresses
- Dangerous attachments
- Bulk-sending patterns

If the server suspects spam, it may:

- Send the message to the **spam folder**,
- Reject it, or
- Quarantine it for review.

The Server Stores It in the User's Mailbox

If the message passes all checks:

- It gets placed into the user's **mailbox folder** on the mail server.
- The mailbox is usually stored in a directory or database system that keeps track of:
 - New messages
 - Read/unread status
 - Attachments
 - Folders (Inbox, Sent, Trash, Spam)

This mailbox waits until the recipient's email app (Gmail, Outlook, phone email app) connects using **IMAP or POP3** to retrieve or sync the message.

20. Mention mailbox storage types (IMAP vs POP3).

1. IMAP (Internet Message Access Protocol)

IMAP stores emails on the server.

With IMAP:

- Your emails stay on the mail server
- You can check your email from multiple devices (phone, laptop, tablet)
- Everything stays synced — if you read or delete a message on one device, it updates everywhere

IMAP is ideal for modern email apps like Gmail, Outlook, and Apple Mail.

Analogy:

IMAP is like visiting the post office to read your letters — you read them there, but the letters stay in the post office.

2. POP3 (Post Office Protocol Version 3)

POP3 downloads emails to your device and usually removes them from the server.

With POP3:

- Your device downloads the message
- The email is removed from the server (in most setups)
- Other devices won't see the same message

POP3 is older and less common today.

Analogy:

POP3 is like the mailman delivering letters to your home — once delivered, the post office no longer has them.

In short:

Feature	IMAP	POP3
Storage	On the server	Downloaded to device
Sync across devices	Yes	No
Best for	Multiple devices	Single device
Modern usage	Very common	Older method

21. Explain IMAP and POP3 as protocols that .

Describe how the message is downloaded or synced from the mailbox to the recipient's email app.

Relate it to a real mailbox where new messages arrive.

IMAP and POP3 Explained

IMAP and POP3 are **email retrieval protocols** — they help your email app (like Gmail app, Outlook, Apple Mail, etc.) get the messages from the mail server.

IMAP (Internet Message Access Protocol)

What it does

- **Syncs email between all your devices.**
- Emails **stay on the server**, and your phone/laptop just shows a copy.
- If you **delete or read** a message on one device, it updates everywhere.

Real-life example

Imagine a **real mailbox outside your home**.

You *open the lid*, look at letters, but **leave them inside**.

You can check them anytime from anywhere — they stay in the mailbox.

How IMAP downloads/syncs

- The email app connects to the server.
- It **syncs** the list of emails.
- When you open a mail, it loads the content from the server.
- Nothing is permanently moved—everything stays stored on the server.

POP3 (Post Office Protocol version 3)

What it does

- **Downloads the email to one device** and usually **deletes it from the server**.
- Emails may not sync across devices.

Real-life example

Imagine opening your mailbox and **taking the letters inside your house**.
Once taken, the mailbox outside becomes **empty**.

How POP3 downloads

- The email app connects to the server.
- It **downloads messages** to your device.
- After download, the server's mailbox often becomes empty (unless you change settings).

How the message reaches your app (IMAP vs POP3)

Using IMAP:

1. Your app connects to the mailbox on the server.
2. It **syncs** all folders (Inbox, Sent, Drafts).
3. Emails stay on the server.
4. You only download the body or attachments when needed.

Using POP3:

1. Your app connects to the server.
2. It **downloads all new emails**.
3. Deletes them from the server (default behavior).
4. Emails now exist only on your device.

Quick Summary Table

Feature	IMAP POP3	
Where emails stay	Server	Local device
Sync across devices	Yes	No
Good for multiple devices	yes	no
Deletes mail from server	No	Yes (by default)

